tf.compat.v1.ConditionalAccumulator Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/ConditionalAccumulator

> *A conditional accumulator for aggregating gradients.*

Function Signatures

```
tf.compat.v1.ConditionalAccumulator( dtype, shape=None,
shared_name=None, name='conditional_accumulator',
reduction_type='MEAN' )
```

Code Examples

```
tf.compat.v1.ConditionalAccumulator(
dtype,
shape=None,
shared_name=None,
name='conditional_accumulator',
reduction_type='MEAN'
)
```

```
tf.compat.v1.ConditionalAccumulator(
dtype,
shape=None,
shared_name=None,
name='conditional_accumulator',
reduction_type='MEAN'
)
```

```
apply_grad(
grad, local_step=0, name=None
)
```


```
apply_grad(
grad, local_step=0, name=None
)
```


```
num_accumulated(
name=None
)
```


```
num_accumulated(
name=None
)
```


```
set_global_step(
new_global_step, name=None
)
```


```
set_global_step(
new_global_step, name=None
)
```

Documentation

A conditional accumulator for aggregating gradients.

Inherits From: ConditionalAccumulatorBase

Up-to-date gradients (i.e., time step at which gradient was computed is equal to the accumulator's time step) are added to the accumulator.

Extraction of the average gradient is blocked until the required number of gradients has been accumulated.

Attributes

Methods

apply_grad

View source

Attempts to apply a gradient to the accumulator.

The attempt is silently dropped if the gradient is stale, i.e., local_step is less than the accumulator's global time step.

num_accumulated

View source

Number of gradients that have currently been aggregated in accumulator.

set_global_step

View source

Sets the global time step of the accumulator.

The operation logs a warning if we attempt to set to a time step that is lower than the accumulator's own time step.

take_grad

View source

Attempts to extract the average gradient from the accumulator.

The operation blocks until sufficient number of gradients have been successfully applied to the accumulator.

Once successful, the following actions are also triggered:

- Counter of accumulated gradients is reset to 0.
- Aggregated gradient is reset to 0 tensor.
- Accumulator's internal time step is incremented by 1.
